

The Al Reem District Problem

Team- Bridges and Bulls

School: Dubai English Speaking School

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Our Solution: Covered Walkway and Small Electric Shuttle

We are proposing a covered walkway along the full 1.6km route from the residential blocks to the bus interchange, with a small electric shuttle running alongside it for people who need it most.

The way we see it, the problem is not that 1.4km is too far to walk. It is that 1.4km is too far to walk when it is 45 degrees outside. So the solution needs to deal with the heat, not the distance. That is why we went with shade over the whole route rather than just a bigger bus system.

We thought about doing just the shuttle or just the walkway but neither one works on its own. A shuttle big enough to carry all 10,800 commuters would need way more vehicles than we can afford to run. A walkway on its own helps but it is still going to be really hot even with shade. It does not solve the problem for people who physically cannot walk that far in the heat, like elderly residents or families with young kids. Combining them means the walkway covers most people and the shuttle is there for the people who genuinely need a ride.

The Walkway

The route follows the existing pedestrian path but we would widen it from about 2m to 5m, half for the shuttle lane and half for walking. Along the full length we would build a continuous canopy with solar panels on top. The solar panels do two things at once. They shade the walkway and they generate electricity that helps power the shuttle.

We looked up how much power solar panels produce in the Gulf and based on what we found, roughly 200W per square metre with about 6 hours of strong sunlight per day, the canopy area of around 4,900 square metres could generate something like 5,000 to 6,000 kWh per day. The shuttle fleet would only use around 300 to 350 kWh per day so there is a big surplus that can go toward powering the stops and the interchange. We are not completely confident in these numbers but the general point is that the solar panels would produce way more than the shuttles need.

We would also put six covered stops along the route, one every 250 metres or so. For cooling at the stops we looked into passive downdraft cooling. Basically it is a chimney structure that pulls hot air up and draws cooler air across a wet mesh at ground level. It can drop the temperature by around 8 to 12 degrees without needing air conditioning. This is not something we invented. It is based on traditional wind tower architecture in the Gulf and there are modern versions of it already working in Masdar City. We think this is better than mechanical AC because it is much cheaper to run and does not break down as easily.

The Shuttle

Four small electric minibuses, 12 seats each. They would loop the route continuously, every 8 minutes during morning and evening commute times, every 15 minutes the rest of the day, running from 5:30 AM to 11 PM. Each vehicle has an 80 kWh battery which from what we could find should last about 14 hours of low speed driving before it needs to charge. Charging happens overnight at a depot next to the bus interchange.

In terms of how many people this actually moves, during the morning peak from 5:30 to 8:30, about 70 percent of the 10,800 commuters travel, so around 7,500 people. Four shuttles carrying 12 people each, doing about 7 to 8 trips per hour over 3 hours, gives us roughly 1,000 to 1,100 rides during that window. So the shuttle handles about a thousand people during peak and the rest walk the shaded route. The afternoon peak works the same way in reverse.

We are not claiming the shuttle replaces walking for everyone. It is there to take the most dangerous trips off foot traffic, mainly people travelling during the hottest part of the day.

Costs

We tried to estimate these based on what we could find online for similar projects in the UAE. Some of these are rough but we have tried to be realistic rather than just picking low numbers to make it look good.

Capital Costs

Route widening and resurfacing, 1.6km: AED 4,800,000

Solar canopy structure and panels, approximately 4,900 square metres: AED 14,700,000

Six passenger stops with passive cooling: AED 3,600,000

Four electric minibuses at about AED 350,000 each: AED 1,400,000

Charging depot and electrical connection: AED 2,200,000

Signage, safety barriers, lighting: AED 1,100,000

Contingency because we have probably got some numbers wrong: AED 2,780,000

Total: AED 30,580,000

This is under the AED 40 million cap. We have about AED 9.4 million left unspent. We did not try to use the full budget because spending money we do not need to spend would weaken the operating model and would not be a responsible use of public funds.

Annual Operating Costs

Shuttle drivers, 8 on rotating shifts: AED 384,000

Vehicle maintenance and battery reserve: AED 280,000

Grid electricity after solar offset: AED 95,000

Stop cleaning, canopy maintenance, cooling upkeep: AED 320,000

Route maintenance: AED 180,000

Insurance and admin: AED 140,000

Contingency: AED 139,900

Total: about AED 1,540,000 per year

This is under the AED 2.5 million annual budget. The surplus of roughly AED 960,000 per year gives us a buffer if costs end up higher than we estimated or if the shuttle fleet needs expanding later.

Who Pays

The municipality pays for everything. We thought about charging a fare for the shuttle but decided against it for a few reasons. The people most affected by the heat problem are the lowest paid workers and a fare just replaces the taxi cost with a different cost, which defeats the purpose. Also collecting fares would need ticket machines or a payment system which adds cost and complexity that is not worth it at this small scale. And the municipality has already set this budget aside so they do not actually need to recover costs from riders.

The return for the municipality is not revenue. It is that bus ridership stops collapsing every summer, fewer workers get heat related health problems, and the workforce arrives at the logistics hub in better condition. We cannot put an exact number on those benefits but they are real.

Cost Per Person

With about 10,800 people using the route daily and annual operating costs of AED 1,540,000, the cost works out to about AED 142 per user per year. That is roughly 39 fils per person per day.

For comparison, workers currently pay AED 15 to 25 per taxi trip. So this solution costs the municipality less than 2 percent of what each worker is currently paying out of their own pocket per trip. That seems like a reasonable deal.

What This Does Not Do Well

We want to be honest about the weaknesses.

Even with shade and misting, walking 1.4km when it is 45 degrees outside is still uncomfortable. The walkway makes it safe but it does not make it pleasant. On the worst days people will still be hot and tired by the time they reach the interchange.

The shuttle is also small. Two vehicles running at once during peak cannot carry everyone who might want a ride. There will probably be queues. We accepted this because a shuttle fleet big enough to carry all 10,800 commuters would need at least 15 vehicles and operating costs way above the annual budget.

Some of our cost estimates are also uncertain. We did our best with what we could find online but we have never costed a construction project before and some of these figures could be off. That is what the contingency is for.

A system that completely eliminates heat exposure for every resident at every time of day would cost roughly three times this budget. We chose the version that helps the most people within the constraint rather than the version that helps fewer people perfectly.

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